

🔑 Key Power Signals to Check on I/O Chip (Super I/O or EC)

Signal Name	Full Form	Expected Voltage	What It Means
+3VPCU / +3VALW	3V Always / 3V Power Control Unit	~3.3V	Standby voltage; must be present always when PSU is plugged
+5VSB	5V Standby	~5V	Standby voltage from PSU; must be present
PWRBTN#	Power Button	~3.3V → 0V when pressed	Goes low (0V) when power button is pressed
RSMRST#	Resume Reset	~3.3V	If LOW → chipset in reset; must be HIGH to proceed
LID_SW#	Lid Switch (Laptop)	~3.3V normally	Goes low when lid is closed; must be high to power on
SUS_PWR	Suspend Power	~3.3V	Enables certain power rails; shows S3/S4/S5 states
SLP_S3#, SLP_S5#	Sleep Signals	~3.3V (high = awake)	If LOW → in sleep or soft-off mode

🖥️ Signals to Check on the Chipset (PCH – Platform Controller Hub)

Signal Name	Full Form	Expected Voltage	What It Tells You
PWR_OK	Power OK from PSU	~3.3V or ~5V	Power is stable and ready; needed before boot starts
PLTRST#	Platform Reset	~3.3V (high = released)	If stuck LOW, platform is still in reset
CLKREQ#	Clock Request	~0V (active low) or toggling	Indicates if the device is requesting clock
ALL_SYS_PWRG D	All System Power Good	~3.3V	All power rails are good; comes from EC or PMIC
SYS_RESET#	System Reset	~3.3V	Must be high for system to run
VCCP / VCC_CORE	CPU Core Voltage	0.8V – 1.2V (approx)	Check if CPU is getting power from VRM

🧠 Minimum Signals for Boot to Start

At **minimum**, to troubleshoot a "no power" or "no boot" board:

Signal	Should Be Present?	Check At
+5VSB	✅ Yes	PSU or standby rail
+3VALW / +3VPCU	✅ Yes	Near EC/SIO
PWRBTN#	✅ Goes LOW when pressed	EC or SIO input pin
RSMRST#	✅ High (3.3V)	SIO or EC output
PLTRST#	✅ High (3.3V)	PCH reset pin
VCC_CORE	✅ 0.8V–1.2V	CPU power rail

Step-by-Step Power Debugging Using Multimeter

1. Plug in PSU (don't press power yet)

-  Check **+5VSB**
-  Check **+3VPCU** (always ON)
-  Check **EC/SIO VCC** (check chip voltage pins)

2. Press power button

-  **PWRBTN#** goes from 3.3V → 0V (momentarily)
-  **RSMRST#** goes HIGH
-  **SIO/EC** sends **PCH_ON#** to turn on main PSU rails

3. Check main rails

-  +12V, +5V, +3.3V come ON
-  **PWR_OK** goes HIGH

4. Check signals to chipset

-  **PLTRST#** goes HIGH
-  **SLP_S5#, SLP_S3#** are HIGH (out of sleep)
-  **VCORE** starts (check near CPU)